LESSON 12
PROBLEM SOLVING STRATEGIES

Overview:

Problem solving has a wide range of applications in any discipline. This is also our tool in decision making. This lesson will assist students applying a problem-solving strategy to all types of word problems.

Learning Outcomes:

At the end of this lesson, the students can:

1. identify the problem-solving strategy to be used in a given problem; and
2. solve problems using the different problem-solving strategies.

Materials Needed:

- Modules (hard and soft copy)
- Devices (laptop and mobile phone)
- Social Media Platforms (Moodle, Google classroom, Facebook, YouTube and etc.).

Duration: 3 hours

Learning Contents:

12.1 PROBLEM SOLVING STRATEGIES FROM GEORGE POLYA

George Polya identified four principles that form the basis for any serious attempt at problem solving:

Step 1. Understand the problem
Step 2. Devise a plan
Step 3. Carry out the plan
Step 4. Look back (reflect)

Step 1. Understand the problem

a) What are you asked to find out or show?
b) Can you draw a picture or diagram to help you understand the problem?
c)Can you restate the problem in your own words?
d)Can you work out some numerical examples that would help make the problem more clear?

Step 2. Devise a plan

A partial list of Problem-Solving Strategies includes:

Guess and check	Draw a diagram
Look for a pattern	Solve a simpler problem
Make an orderly list	Make a Table
Use logical reasoning	Work backwards

Step 3. Carry out the plan

a) Carrying out the plan is usually easier than devising the plan

b) Be patient – most problems are not solved quickly nor on the first attempt

c) If a plan does not work immediately, be persistent

d) Do not let yourself get discouraged

e) If one strategy isn't working, try a different one

Step 4. Look back (reflect)

a) Does your answer make sense? Did you answer all of the questions?

b) What did you learn by doing this?

c) Could you have done this problem another way – maybe even an easier way?

EXAMPLE: Mr. Jones has a total of 25 chickens and cows on his farm. How many of each does he have if all together there are 76 feet? (Guess and Check.)

Solution:

Step 1: Understanding the problem

We are given in the problem that there are 25 chickens and cows. All together there are 76 feet. Chickens have 2 feet and cows have 4 feet. We are trying to determine how many cows and how many chickens Mr. Jones has on his farm.

Step 2: Devise a plan

We are going to use Guess and Check.

Step 3: Carry out the plan:

number of chickens	number of cows	number of chickens' feet	number of cows' feet	total number of feet
20	5	40	20	60
21	4	42	16	58

Notice that we are going at the wrong direction since the number of feet decreases as we increase the number of chickens. Let us try to reduce the number of chickens.

number of chickens	number of cows	number of chickens' feet	number of cows' feet	total number of feet
16	9	32	36	68
14	11	28	44	72

Step 4: Looking back:

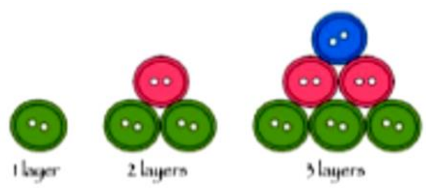
Check:

$12 + 13 = 25$ heads

$24 + 52 = 76$ feet

We have found the solution to this problem. We could use this strategy when there are a limited number of possible answers and when two items are the same but they have one characteristic that is different.

EXAMPLE: Edith arranged buttons in a triangular shape as shown. How many buttons will there be in a triangle that has 12 layers? (Look for a pattern.)



Solution:

It is noted that the buttons were arranged in a triangular shape. We are asked to determine the number of buttons be needed to form a triangle having twelve (12) layers.

Observe that in a single layer, there is only one button; for two layers, there are $2 + 1 = 3$ buttons; for three layers, there are $3 + 2 + 1 = 6$. From this, we can predict that if there are 4 layers, the number of buttons will be$4 + 3 + 2 + 1 = 10$; for 5 layers, there will be $5 + 4 + 3 + 2 + 1 = 15$ buttons. Continuing this pattern, a 12-layer triangle has $12 + 11 + 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 78$ buttons.

EXAMPLE: Find the median of the following test scores:

$73, 65, 82, 78, \text{and } 93.$

Solution:

Make an orderly list from smallest to largest:

65, 73, 78, 82, 93

Since 78 is the middle number, the median is 78.

EXAMPLE: How many diagonals does a 13-gon have? (Make a table)

Solution:

Make a table:

Number of sides	Number of diagonals
3	0
4	2
5	5
6	9
7	14
8	20

Look for a pattern. Hint: If n is the number of sides, then $\frac{n(n-3)}{2}$ is the number of diagonals. Explain in words why this works. A 13-gon would have $\frac{13(13-3)}{2} = 65$ diagonals.

EXAMPLE: Karen is thinking of a number. If you double it, and subtract 7, you obtain 11. What is Karen's number? (Work Backwards)

Solution:

We start with 11 and work backwards.

The opposite of subtraction is addition. We will add 7 to 11. We are now at 18.

The opposite of doubling something is dividing by 2. $18 \div 2 = 9$

This should be our answer. Looking back:

$9 \times 2 = 18 - 7 = 11$

Hence, Karen's number is 9.

Learning Activity:

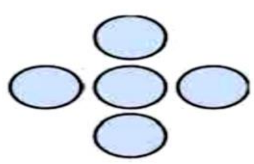
Think-Pair-Share. Solve the following problems using any two problem solving strategies. Student A must choose a problem-solving strategy different from Student B. Afterwards, they will compare their work and check pair's work.

1) Beryl's car can travel at a top speed of 55 miles per hour on the highway. Her sister Sue's car can only travel 45 miles per hour on the highway. If the Sue leaves two hours early, how many hours will it take for Beryl to catch up to Sue?
2) Irene was given 48 stones. She wanted to arrange them into a rectangle in her garden. How many different rectangular arrangements are possible using 48 stones?
3) Christina is thinking of a number. If you multiply her number by 93, add 6, and divide by 3, you obtain 436. What is her number?
4) Find the next 2 numbers in the sequence 1, 4, 7, 10, 13….

Learning Evaluation:

Solve the following problems using any of the problem-solving strategies.

1) 1. How many different amounts of money can you have in your pocket if you have only three coins including only 5-peso coin and 25-centavo coin?
2) Can perfect squares end in a 2 or a 3?
3) In the sequence:1, 2, 4, 8 … Find the next two numbers.
4) Place the digits 8, 10, 11, 12, and 13 in the circles to make the sums across and vertically equal 31.



References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.

Polya, G. (1957). *How to Solve It*. 2nd Edition. Princeton University Press.